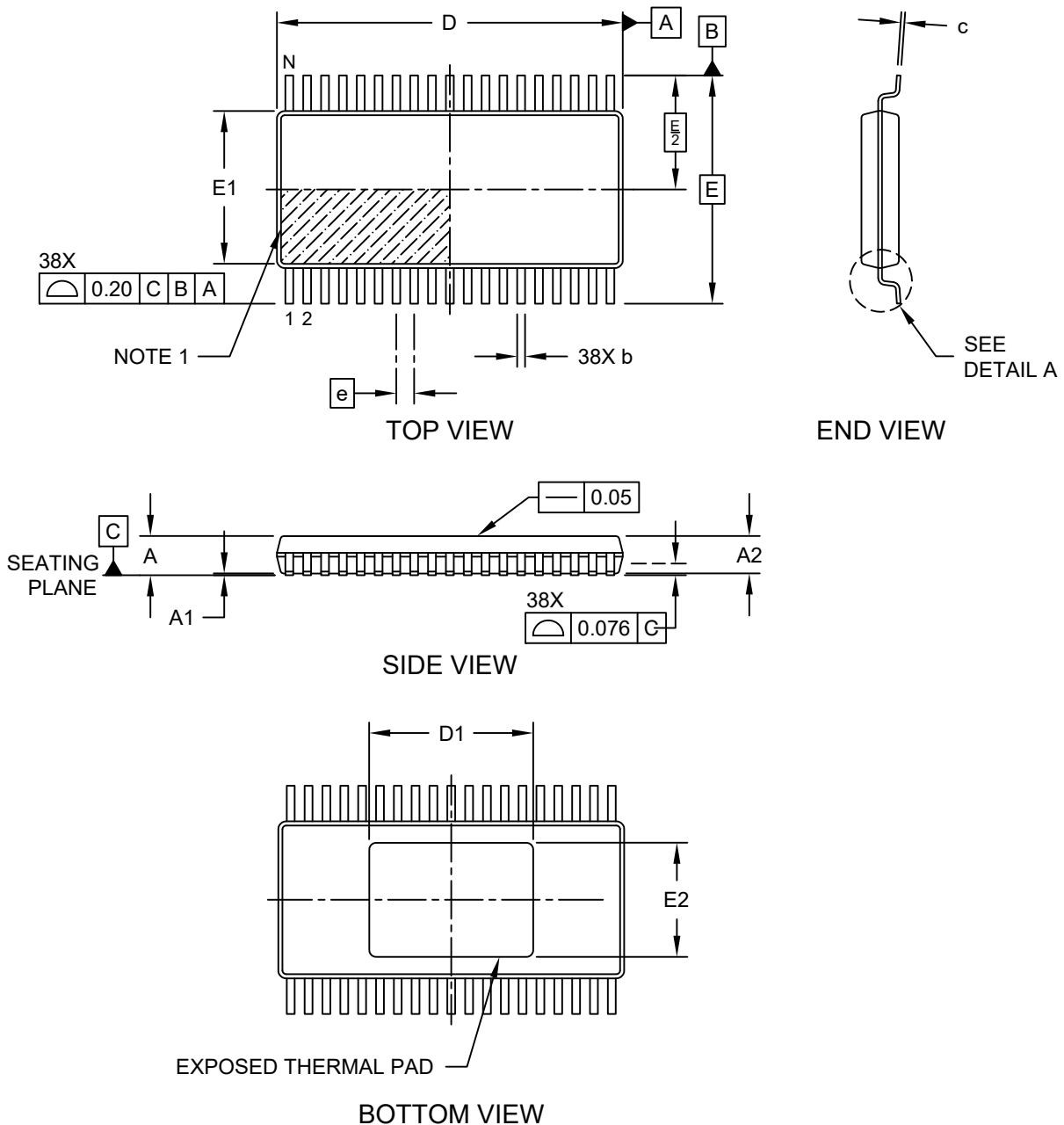


Packaging Diagrams and Parameters

38-Lead Thin Shrink Small Outline Package (SBX) - 4.4 mm Body [TSSOP] With 4.6x 3.2 mm Exposed Pad

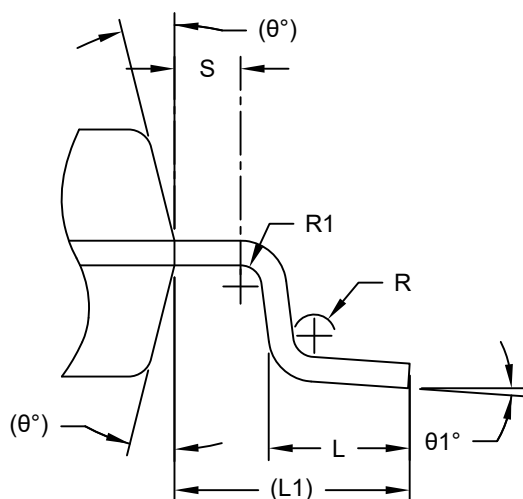
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



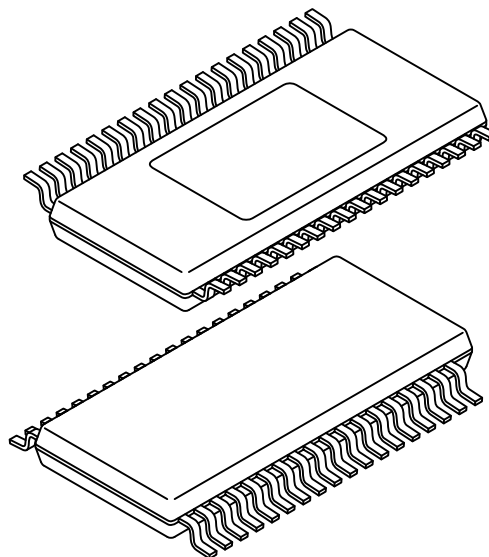
Packaging Diagrams and Parameters

38-Lead Thin Shrink Small Outline Package (SBX) - 4.4 mm Body [TSSOP] With 4.6x 3.2 mm Exposed Pad

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DETAIL A



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	38		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	1.10
Standoff	A1	0.05	-	0.15
Terminal Thickness	A2	0.85	0.90	0.95
Overall Length	D	9.60	9.70	9.80
Exposed Pad Length	D1	4.50	4.60	4.70
Overall Width	E	6.40 BSC		
Molded Package Width	E1	4.30	4.40	4.60
Exposed Pad Width	E2	3.10	3.20	3.30
Terminal Width	b	0.17	-	0.27
Terminal Width	c	0.09	-	0.20
Terminal Length	L	0.50	0.60	0.70
Terminal Length	L1	1.00 REF		
Lead Shoulder	S	0.20	-	-
Terminal Foot Angle	θ1	0°	-	8°
Mold Draft Angle	θ	14° REF		

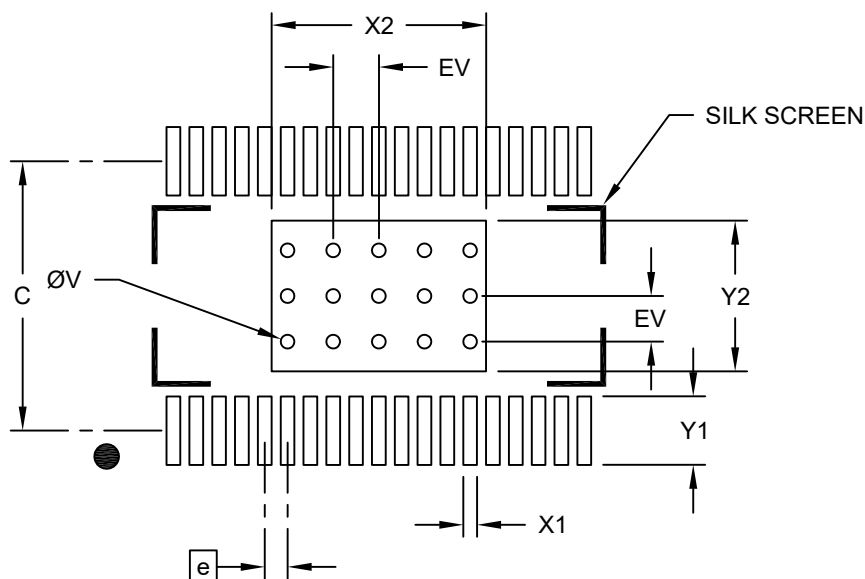
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

38-Lead Thin Shrink Small Outline Package (SBX) - 4.4 mm Body [TSSOP] With 4.6x 3.2 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Center Pad Width	X2			4.70
Center Pad Length	Y2			3.30
Contact Pad Spacing	C		5.90	
Contact Pad Width (X38)	X1			0.30
Contact Pad Length (X38)	Y1			1.50
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	EV		1.00	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process